MIL-T-36308A
6 March 1975
SUPERSEDES
MIL-T-36308 (DM)
19 October 1964

MILITARY SPECIFICATION

TEST KIT, ALKALINE PHOSPHATASE LEVEL

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers a Test Kit, Alkaline Phosphatase Level, suitable for use in the rapid, semi-quantitative determination of alkaline phosphatase level of human plasma or serum. (See 6.2)

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

PPP-B-566	Box, Folding, Paperboard.
PPP-B-636	Box, Fiberboard.
PPP-B-676	Box, Setup.
PPP-C-186	Containers, Packaging and Packing for Drugs, Chemicals and Pharmaceuticals.

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

FSC 6505

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply.

AMERICAN PHARMACEUTICAL ASSOCIATION PUBLICATION

National Formulary.

(Application for copies should be addressed to the Mack Publishing Company, Easton, PA 18042.)

U. S. DEPARTMENT OF HEALTH, EDUCATION, AND WELFARE, FOOD AND DRUG ADMINISTRATION

Federal Food, Drug, and Cosmetic Act and Regulations Promulgated Thereunder.

(Application for copies should be addressed to the Food and Drug Administration, Washington, DC 20204.)

U.S. PHARMACOPEIAL CONVENTION, INC.

Pharmacopeia of the United States

(Application for copies should be addressed to the Mack Publishing Company, Easton, PA 18042.)

3. REQUIREMENTS

3.1 Description. Shall be a test kit suitable for use in the rapid, semi-quantitative determination of alkaline phosphatase level of human plasma or serum.

3.1.1 Components. The test kit shall consist of the following components: (suitable for 48 determinations)

Color Chart	1
Test tube rack	1
Sodium hydroxide solution, 0.5N	4cc
Sodium phenolphthalein phosphate test tablets, 0.3 mg (buffered)	48
Test tube, 0.5 cc capacity	48

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3.1.1.1 Color chart. Shall be a chart with color chips, firmly affixed on paper of suitable quality, designating the colors produced in the normal, borderline, elevated, and highly elevated readings from the serums of nonjaundiced, latent jaundiced, and jaundiced patients. Color chips shall match the colors produced in the standards specified in 4.3.1. In addition, the color chart shall include directions for use, interpretation of test and any other information deemed appropriate to obtain maximum use of the product.

3.1.1.2 Test tube rack. Shall be a setup tray with die-cut platform and shall contain 8 holes for holding the test tubes while making tests. The rack shall be fabricated from mounting board material sufficiently strong for holding the test tubes in an upright position.

3.1.1.3 Sodium hydroxide solution. Shall be 4.0 cc of 0.5N sodium hydroxide. The solution shall assay to contain not less than 1.90 grams and not more than 2.10 Grams of sodium hydroxide per 100 ml when titrated with 0.5N HCl using phenolphthalein T. S. as the indicator.

3.1.1.4 Test tablets.

3.1.1.4.1 Composition. Shall be buffered tablets containing 0.3 mg sodium phenolphthalein phosphate in each tablet.

3.1.1.4.2 Hardness. The hardness of tablets shall be such as to prevent breakage, crumbling, or powdering of the tablets in the course of ordinary handling and storage.

3.1.1.4.3 Free phenolphthalein. Tablets shall contain not more than 13.5 mcg of free phenolphthalein for each 15 mg of sodium phenolphthalein phosphate, when tested in accordance with 4.5.2.

3.1.1.4.4 Color activity. When tested at each standard alkaline phosphatase level (3.0, 6.0, 12.0, and 22.0 Bodansky units), the tablets shall produce the color corresponding to the standard color shown on the color chart.

3.2 Potency period (Expiration dating period). The potency period (expiration dating period) shall be not less than 36 months.

3.3 Delivery. Not more than six (6) months of the expiration dating period shall have elapsed at time of delivery to the Government.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Supplier responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at ant time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4.1.2 No company supplying any ingredient(s) to the contractor will be considered an acceptable facility for the performance of any inspection requirements specified herein.

4.2 Lot. For purposes of this specification, a lot, batch, or control is that single, uniform, and homogeneous quantity of reagent and tablets produced from one formulation, subjected to the same compounding and manufacturing operation, and filled into final containers.

4.3 Inspection. Inspection shall be conducted as necessary, to determine compliance with the requirements of this specification.

4.4 Sampling. Sampling shall be conducted in accordance with the procedures set forth in MIL-STD-105, with an acceptable quality level (AQL) of 1.0 percent defective for major defects and 2.5 percent defective for minor defects.

4.5 Tests.

4.5.1 Color chips. Prepare serum standards containing alkaline phosphatase levels of 3.0, 6.0, 12.0, and 22.0 Bodansky units having an Icterus index of 5. Test each serum standard in accordance with the directions for use for the kit. Compare the colors produced in the sera with the colors of the respective color chips.

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4.5.2 Assay for Free Phenolphthalein.

Reagents:

0.2M Glycine Buffer pH 10.4: Dissolve 15 Gram glycine in 900 ml of distilled water. Adjust pH with 1N NaOH to 10.4 ± 0.1 and dilute to 1 liter with distilled water. Mix thoroughly.

Phenolphthalein Standard Solution: Dissolve 100 mg of phenolphthalein in sufficient Alcohol USP to make 100 ml (1 mg/ml). Mix thoroughly and keep tightly stoppered.

Phenolphthalein Working Standard Solution: Pipet 1 ml Phenolphthalein Standard Solution into a 100-ml flask, dilute to volume with 0.2M glycine buffer, pH 10.4, and mix thoroughly (Solution A). Pipet 1.0 ml of Solution A into a 10-ml volumetric flask. Dilute to volume with 0.2M glycine buffer pH 10.4 and mix thoroughly (10.0 mcg/10 ml).

Procedure:

Dissolve 50 tablets in sufficient 0.2M glycine buffer pH 10.4 to make exactly 10 ml, mix thoroughly and filter solution through Whatman No. 1 filter paper. Determine the maximum absorbance of sample solution and Phenolphthalein Working Standard Solution (10.0 mcg/10 ml) in a suitable spectrophotometer at a wavelength of approximately 550 mu, using 1 cm cells. Set the instrument at zero absorbance with 0.2M glycine buffer pH 10.4.

4.6 Examination for preparation for delivery. An examination shall be made to determine whether the packaging, packing, and marking comply with the requirements of Section 5. Defects shall be scored as specified in Table I. Sampling shall be in accordance with MIL-STD-105. The sample unit shall be one container fully prepared for delivery. The lot shall be the number of containers offered for inspection at one time. The inspection level shall be S-2 and the acceptable quality level shall be 4.0 defects per hundred units.

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Table I. Examination for Preparation for Delivery

Examination	Defect
Containers	Not as specified.
Quantity	Quantity in interior packages and shipping containers not as specified.
Materials	Any component missing or damaged.
Workmanship	Inadequate application of components, such as incomplete closure of container flaps, tape improperly applied, or con-containers distorted.
Markings	Omitted, incorrect, illegible, improper size, location, sequence, or method of application.

5. PREPARATION FOR DELIVERY

5.1 Preparation for Delivery shall be in accordance with all applicable requirements of PPP-C-186 and as specified herein.

5.2 Packaging. Packaging shall be level A or C (as specified). (See 6.1)

5.2.1 Level A.

5.2.1.1 Immediate container. The immediate container for the tablets and Sodium Hydroxide Solution shall be a commercially available immediate container considered acceptable by the Food and Drug Administration (FDA) for the product contained therein and shall be of such compatibility that neither the contents nor the container are altered in any way by each other.

5.2.1.2 Unit package. Each test kit, consisting of a color chart, test tubes, test tube rack, and the immediate containers containing the tablets and solution, shall be packaged in a snug-fitting box constructed in accordance with PPP-B-566, PPP-B-636, class domestic or PPP-B-676. The unit package shall be designed with suitable interior fittings or shall contain sufficient cushioning material to insure against shifting of or damage to the contents. Closure of the box shall be secured to prevent accidental opening.

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5.2.1.3 Intermediate package. Twelve unit packages shall be packaged in a snug-fitting box constructed in accordance with PPP-B-636, type CF, class domestic and having a minimum bursting strength test of 200 pounds. Closure of the box shall be secured to prevent accidental opening.

5.2.2 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.3 Packing. Packing shall be level A or C as specified (see 6.1).

5.3.1 Level A. Eight intermediate packages shall be packed in an exterior container designed for a type 2 load as specified in paragraph 5.4.4 of PPP-C-186.

5.3.2 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.4 Marking.

5.4.1 Labeling. Labeling shall be in accordance with the requirements of the Federal Food, Drug and Cosmetic Act, and shall include the information required below:

5.4.1.1 Immediate containers. Each immediate container label shall bear the following information:

(a) label information in accordance with commercial practice.

(b) the expiration date.

5.4.1.2 Unit packages. Each unit package label shall bear the following information:

(a) labeling information in accordance with commercial practice.

(b) the National Stock Number.

NOTES: (1) the National Stock Number shall not be preceded by any prefix.
(2) Strip labeling may be used.
(3) Label adhesion test shall not apply.

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5.4.2 Intermediate package. Each intermediate package shall be marked in accordance with MIL-STD-129. Lot (control) number, contract or purchase order number and name of contractor shall be shown. Type I shelf life marking as specified in MIL-STD-129 shall also be shown. Marking shall include the legends:

"DO NOT PERMIT TO FREEZE"
and
"STORE BETWEEN 2° - 8°C (35° - 46°F)".

5.4.3 Exterior container. Exterior containers shall be marked in accordance with MIL-STD-129. Lot (control) number shall be shown. Type I shelf life marking as specified in MIL-STD-129 shall also be shown. Marking shall include the legends:

"DO NOT PERMIT TO FREEZE"
and
"STORE BETWEEN 2° - 8°C (35° - 46°F)".

NOTE: The unrefrigerated shipping time for this item shall be not more than 7 days (168 hours).

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Number and date of this specification.
(b) the National Stock Number.
(c) Selection of applicable levels of packaging and packing (see 5.2 and 5.3).

6.2 This specification covers the following item:

National Stock No.	Item Identification
6505-00-890-1092	TEST KIT, ALKALINE PHOSPHATASE LEVEL, 48 Tests

6.3 This specification does not cover all tpyes, classes, grades, or sizes of the commodity indicated by the title of this specification, or those which are commercially available, but is intended to cover the types which are normally procured to meet Military requirements.

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6.4 All reference to PPP-C-186 refers to Interim Federal Specification PPP-C-00186 which is mandatory for use until superseded by a revision.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

Project No. 6505-1380
is

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INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. *DoD contractors, government activities, or manufacturers/ vendors who are prospective suppliers of the product* are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

MIL-T-36308A

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

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